

Coding & Solution

Date	05 July 2026
Team ID	xxxxxx
Project Name	HDI Predictor (Human Development Index Prediction Using Machine Learning)
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/ramyamamidisetti
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask

Key Features Implemented	Data preprocessing, Machine Learning model training, HDI prediction, Flask web application, User input form, Real-time prediction
Pending / Incomplete Features	User authentication, Database integration, Prediction history, Graphical dashboard
Setup / Run Instructions	Install required libraries using <code>pip install -r requirements.txt</code> , run <code>python app.py</code> , open <code>http://127.0.0.1:5000</code> in a web browser.



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	No

Additional Notes / Comments:

The project successfully predicts the Human Development Index (HDI) using Machine Learning. It is developed using Python, Flask, Scikit-learn, Pandas, HTML, and CSS. The application provides a simple user interface for entering development indicators and instantly displays the predicted HDI value. The code follows a modular structure and can be extended with database support, authentication, and advanced visualizations in future versions.